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Artificial intelligence

Presenter: Frank Wallace, Retired software engineer

Prepared for ORT America

November 4, 2026

A short family of slides that sets the context for the vocabulary work. Two threads run through it: the alphabet soup of AI terms, and the much older question of what intelligence is and how we measure it. Binet's test in 1905 and today's AI benchmarks turn out to be the same idea, a century apart.

ARTIFICIAL INTELLIGENCE, AS DICKENS MIGHT SEE IT

"It was the best of times, it was the worst of times..."

"...the period was so far like the present period, that some of its noisiest authorities insisted on its being received, for good or for evil, in the superlative degree of comparison only."

— Charles Dickens, *A Tale of Two Cities* (1859)

AI carries real promise and real peril. The wisdom lies in resisting the superlatives on both sides.

This is the opening sentence of A Tale of Two Cities, published in 1859 and set in 1775, on the eve of the French Revolution, in London and Paris. Dickens, writing about eighty years later, says that era was much like his own. The full sentence pairs opposites: the age of wisdom and of foolishness, the spring of hope and the winter of despair, everything before us and nothing before us. Its last clause is the sly part: he gently mocks the "noisiest authorities" who could only describe their own times in extremes. Point for AI: the stakes really do run in both directions, and that is exactly why the hype on either side, utopia or catastrophe, deserves resistance. How AI turns out depends on careful, unglamorous choices people make.

A familiar face?



A WORD ON METHOD

Eating Our Own Dog Food



1970s A TV pitch

Lorne Greene, Bonanza's Ben Cartwright, told Alpo viewers he fed it to his own dogs. Probably the origin, though not proven.

1988 Tech adopts it

Microsoft's Paul Maritz sent a memo titled “Eating our own Dogfood,” and the phrase spread across the tech industry.

Today This talk

This presentation is dog food too: I used AI at every step, from research and drafting to building these slides.

Sources: Wikipedia, “Eating your own dog food”; GeekWire, 2025 interview with Paul Maritz

Before we start, a word about how this talk was made.

"Dogfooding" means using your own product yourself, the way a customer would, before you ask anyone else to trust it.

The usual origin story goes back to the 1970s Alpo dog food commercials. Lorne Greene, who played Ben Cartwright on Bonanza, told viewers Alpo was so good he fed it to his own dogs. I'll be honest that this is a bit speculative: it's the most common explanation, but nobody has proven it. A rival story credits the president of Kal Kan, who was said to eat a can of his own dog food at shareholder meetings.

What isn't speculative is how the tech industry picked it up. In 1988, Microsoft manager Paul Maritz sent an email titled "Eating our own Dogfood," pushing his team to use their own networking product internally. A test server got named "dogfood," and the phrase spread from Microsoft across the industry.

And that brings me to this talk. This presentation is itself an example of dogfooding: I used AI at every step of preparing it, from the research and the drafting to building these slides. So as we go, you're seeing both the subject and the method.

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AI vocabulary · the big picture

What do we mean by “intelligence”?

From narrow AI to “SI,” and from Binet’s first intelligence test to today’s AI benchmarks.

ANI AI AGI ASI SI

A short family of slides that sets the context for the vocabulary work. Two threads run through it: the alphabet soup of AI terms, and the much older question of what intelligence is and how we measure it. Binet’s test in 1905 and today’s AI benchmarks turn out to be the same idea, a century apart.

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The AI family

AI — **Artificial intelligence.** The umbrella term: machines doing things that normally take human intelligence. Coined in 1955 for the Dartmouth research proposal.

ANI

Artificial narrow intelligence (“narrow AI”)

Superb at one job: chess engines, spam filters, face unlock.

Here today

AGI

Artificial general intelligence

As capable as a person at most mental tasks, not just one.

Hypothetical · hotly debated

ASI

Artificial superintelligence

Far beyond the best humans at nearly everything.

Hypothetical

Read the three cards left to right as a ladder: narrow, then general, then super. Only the first rung exists today. Today’s chatbots sit somewhere between ANI and AGI: broad, but uneven. AI is the umbrella that covers all of them. John McCarthy coined “artificial intelligence” in the 1955 proposal for the 1956 Dartmouth summer workshop.

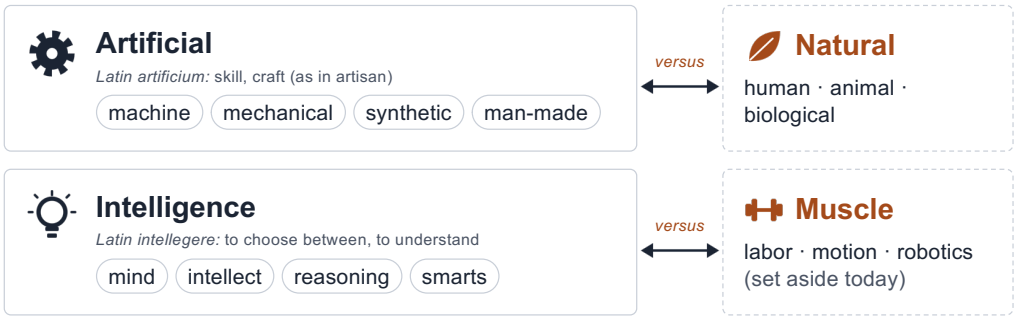
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The AI family

SI **“Super Intelligence.”** President Trump’s new official U.S. name for AI (UN speech, Sept. 22, 2026). A new *label* for today’s AI, not a new *level*, and easy to confuse with ASI.

SI: On Sept. 22, 2026, at the UN General Assembly, President Trump said U.S. government documents would use “super intelligence,” or SI, instead of artificial intelligence, arguing that “artificial” makes the technology sound fake. The State Department told diplomats to use SI the same day. Note the potential confusion: in the field, “superintelligence” (ASI) means a hypothetical AI beyond human ability. SI renames today’s AI; it doesn’t claim we’ve reached ASI. China’s foreign ministry said it “respects” the U.S. choice but kept using “artificial intelligence.”

Unpacking “artificial intelligence”



“Artificial” can also mean *fake*: the very objection behind renaming AI “SI.”

Two words, each defined by what it’s not. Read each row left to right: the word, its Latin root and near-synonyms, then what it’s contrasted with. Artificial versus natural: human, animal, biological. Near-synonyms are machine, mechanical, synthetic, man-made. Its root is the same as “artisan” and “artifice”: something made with skill. Over time it also picked up a second meaning, fake or phony, which is exactly the objection President Trump raised when he renamed AI “SI.” Intelligence versus muscle: thinking, as opposed to labor and motion. Machines that move and lift are robots, which amplify physical power. We’re setting that branch aside and sticking to the thinking side. Latin intellegere literally combines “inter” (between) and “legere” (to choose, to read): to choose between things, to understand.

“AI”: an imperfect name that stuck

John McCarthy, 1955

Picked partly to keep the new field apart from Norbert Wiener’s cybernetics.

Rivals included “complex information processing” and “machine intelligence.”

Science is full of them

Big Bang	Not an explosion in space. A 1993 contest found no better name.
Oxygen	Means “acid-former,” yet plenty of acids contain none.
Positive and negative charge	Franklin guessed in the 1740s, so electric current “flows” backwards.

Imperfect names stick. We learn what they mean, and move on.

A quick slide, under a minute. Many people feel “artificial intelligence” wasn’t the best name: “artificial” can sound fake, and “intelligence” invites us to picture a mind like ours. But that’s normal. Names usually get chosen early, before anyone fully understands the thing, and then they stick. John McCarthy coined the term in the 1955 proposal for the 1956 Dartmouth workshop. He later said he chose it partly to avoid tying the new field to cybernetics and to Norbert Wiener, so the name was partly a turf decision. Herbert Simon and Allen Newell long preferred “complex information processing”; in Britain, “machine intelligence” was common. Modern quips: Ted Chiang suggested “applied statistics,” and Stefano Quintarelli proposed “SALAMI.” Big Bang: Fred Hoyle’s offhand coinage on BBC radio in 1949. Sky & Telescope ran a renaming contest in 1993 and the judges found nothing better. Oxygen: Lavoisier named it “acid-former,” believing all acids contained it; Humphry Davy showed hydrochloric acid doesn’t. Charge: Franklin chose the signs in the 1740s, long before the electron was discovered in 1897, so electrons actually flow opposite to “conventional current.” Tie-in: the 2026 push to rename AI “SI” is the latest attempt to change the name. Whether it catches on outside government remains to be seen.

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Another reading of “AI”: alien intelligence

Historian Yuval Noah Harari suggests the A could stand for *alien*: not from space, but a mind unlike ours.

	Human intelligence	Alien intelligence
Grew from	A body, a childhood, a lifetime	Oceans of text and data
Knows	A lot about a few things	A little about almost everything
When unsure	Hesitates, hedges, frowns	Often sounds just as sure
Mistakes	Familiar, human ones	Strange ones we don’t expect

Not better or worse. Different, and we should expect it to be.

Optional slide, best for the January intro class. The usual contrast is artificial versus natural. Here’s a more modern reading of the A. Yuval Noah Harari (Sapiens, Nexus) has argued that AI is better thought of as alien intelligence: not little green men, but a kind of nonorganic intelligence that doesn’t work like ours. At the CogX Festival in London in September 2023 he called it the first tool in history that can make decisions by itself. The point for us isn’t fear. It’s calibration. Because it didn’t grow up the way we did, it can be dazzling at things we find hard and trip over things a child finds easy, and its errors won’t look like human errors. That leads straight into the next idea: we can’t read its confidence the way we read a person’s.

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“ARTIFICIAL INTELLIGENCE”: JOHN MCCARTHY, 1955

Names it could have had

Alternative term	Who championed it
Complex information processing	Herbert Simon and Allen Newell, two of AI's founders
Machine intelligence	The British preference; Donald Michie's Edinburgh workshops
Cybernetics	Norbert Wiener's broader field, which McCarthy chose to sidestep

John McCarthy coined "artificial intelligence" in the 1955 proposal for the 1956 Dartmouth workshop. He later said he picked it partly to avoid tying the new field to cybernetics and to Norbert Wiener, so the name was partly a turf decision. Simon and Newell long preferred "complex information processing." In Britain, "machine intelligence" was common, notably in Donald Michie's Machine Intelligence workshop series at Edinburgh from 1965.

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AGI AND ASI

Names for what comes next

Term	Origin and emphasis
Strong AI	Philosopher John Searle, 1980: a claim about minds, not a capability level
Human-level AI	The older phrasing of John McCarthy and Nils Nilsson
Transformative AI	Open Philanthropy: defined by impact on the world, not by “intelligence”
Powerful AI	Anthropic's Dario Amodei, who finds “AGI” vague and loaded

“AGI” spread in the 2000s through Ben Goertzel and colleagues; “superintelligence” was popularized by Nick Bostrom's 2014 book.

Each alternative puts the emphasis somewhere different. Strong AI started as John Searle's 1980 term for the philosophical claim that a suitably programmed computer would literally have a mind; it was never meant as a capability threshold. Human-level AI was the older phrasing used by McCarthy and Nilsson. Transformative AI, from Open Philanthropy, sidesteps the word intelligence and asks instead whether the technology changes the world on the scale of the industrial revolution. Dario Amodei of Anthropic uses "powerful AI" and has written that he finds AGI an imprecise term carrying a lot of baggage. AGI as a label spread in the 2000s via Ben Goertzel's circle; superintelligence was popularized by Nick Bostrom's 2014 book of that name.

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Measuring intelligence: Binet's test

1905 Alfred Binet and Théodore Simon (Paris) build a test to spot children who need extra help in school. Binet believed teaching could raise intelligence.

1912 William Stern coins the "intelligence quotient": mental age ÷ actual age × 100.

1916 Lewis Terman's Stanford-Binet brings IQ to America.

Later Misused to rank and label people.
Binet had warned that a test only samples a mind.

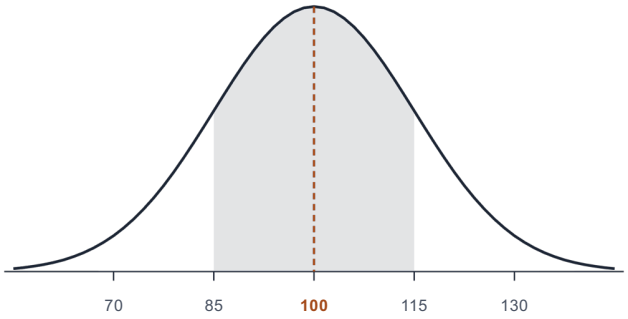


Binet was asked by the French education authorities to find a fair, objective way to identify children who needed extra help, instead of relying on a teacher's impressions. His 1905 test, with Théodore Simon, had about 30 tasks of increasing difficulty. His hopeful twist: he believed intelligence wasn't fixed and that the right teaching could help those children climb. Binet himself never computed an IQ. William Stern proposed the quotient in 1912, and Lewis Terman's 1916 Stanford-Binet made it famous in the U.S. Modern tests are scaled so 100 is the average and about two-thirds of people land between 85 and 115. The dark chapter: the eugenics movement used these tests to label people "feeble-minded," the opposite of Binet's purpose. He had cautioned that a test only samples a person's abilities.

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Measuring intelligence: Binet's test



100 is average by design; about 2 in 3 people score 85–115.

And Einstein? He never took an IQ test. The famous "160" is a guess.

Einstein is everyone's icon of genius, but he never took an IQ test; the figures you see online are after-the-fact estimates.

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Two questions: how much, and at what?

How much?

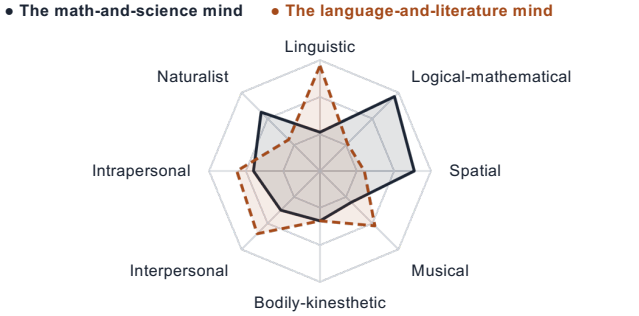
One scale. Most people: 85–115.



One general “g,” or many?
That’s the G in AGI.

At what?

Gardner’s eight intelligences, 1983



Intelligence has two dimensions. How much: the single IQ line. At what: the kind of intelligence. The chart on the right is the visual. Two people, both “smart,” with the same amount of area but very different shapes. The profiles are illustrative, not real scores. Personal story: my school’s program for the “smart kids” treated smart as one thing. I had an aptitude for math and science (the blue shape), not so much for language and history, while a classmate couldn’t get enough of literature and language and didn’t care much for science (the orange shape). We were both “smart,” at different things. Gardner proposed seven intelligences in Frames of Mind (1983) and added naturalist in the 1990s. Many psychologists push back with Charles Spearman’s “g” (1904): people who do well on one kind of test tend to do well on others, which suggests a general factor. That tension, one general ability versus many specific ones, is exactly the “G” in AGI.

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Two questions: how much, and at what?

Animals are specialists, too

Dogs: read our pointing

Chimps: beat students at number memory

Pigs: play joystick video games

Animals: it's tempting to line them up on one scale from cats to gorillas, but each species is smart at what its life demands. Dogs read a human's pointing finger better than chimpanzees do (Hare et al., Science, 2002). A young chimp, Ayumu, beat university students at a flash number-memory task (Inoue and Matsuzawa, Current Biology, 2007). Pigs learned to steer a joystick to play a simple video game (Croney and Boysen, Frontiers in Psychology, 2021).

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From IQ tests to AI benchmarks

Grown, not built

Nobody writes an AI's skills line by line. They emerge during training.

So, like Binet with schoolchildren, we find out what a model can do by testing it.

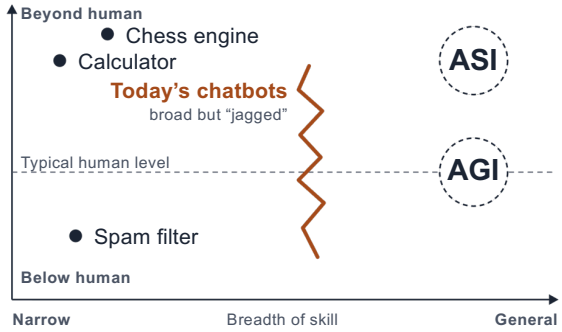
Benchmark	What it tests	Like a…
Humanity's Last Exam	2,500 expert questions across many subjects	Final exam
ARC-AGI	Visual puzzles that are easy for people, hard for AI	Puzzle book
SWE-bench	Fixing real bugs in open-source software	On-the-job test
GPQA Diamond	Graduate science that resists web search	Ph.D. qualifier

Binet's idea, probe a mind with tasks to see where it lands, has turned into what the AI world calls benchmarks. Why we need them: modern AI models are grown more than built. Engineers set up the training, but nobody programs the individual skills, so a model's strengths and weaknesses aren't known until it's tested and probed, much like a child's aptitudes. (Anthropic's CEO Dario Amodei has made this "grown, not built" point.) The four on the slide are among the most-watched in 2026. Humanity's Last Exam was launched in January 2025 by Scale AI and the Center for AI Safety. ARC-AGI is from the ARC Prize Foundation; its newest version, ARC-AGI-3, launched in March 2026 as an interactive game-like test and put frontier models back near zero.

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Where AI fits: how much, and how broad



Narrow AI can beat any human, at one thing.

Today's chatbots are broad but uneven: brilliant at some tasks, weak at others.

AGI and ASI are the dashed circles: not here yet.

SI renames the whole map. It doesn't move anything on it.

This pulls the two threads together. Across: how broad an intelligence is, narrow to general. Up: how much, below human to beyond human. Narrow AI already lives in the upper left: a calculator or a chess engine is superhuman at exactly one thing. A spam filter is narrow and modest. Today's chatbots are much broader, but their skills are "jagged," a term researchers use for AI that can be expert-level at one task and stumble on a nearby one. That's why benchmarks matter: they're how we map the jagged edge. AGI would sit at the right, roughly at human level across the board; ASI above it. Both are still hypothetical, which is why they're drawn dashed. And SI? It's a new official U.S. name for all of this, not a new spot on the map. The map is a sketch to aid thinking, not a measurement.

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Confident is not the same as correct

No face to read

With people, we judge confidence by tone, pauses, and body language. AI often sounds just as sure when it's wrong.

"The computer said so"

Automation bias: trusting a machine over our own judgment. It is much older than AI.

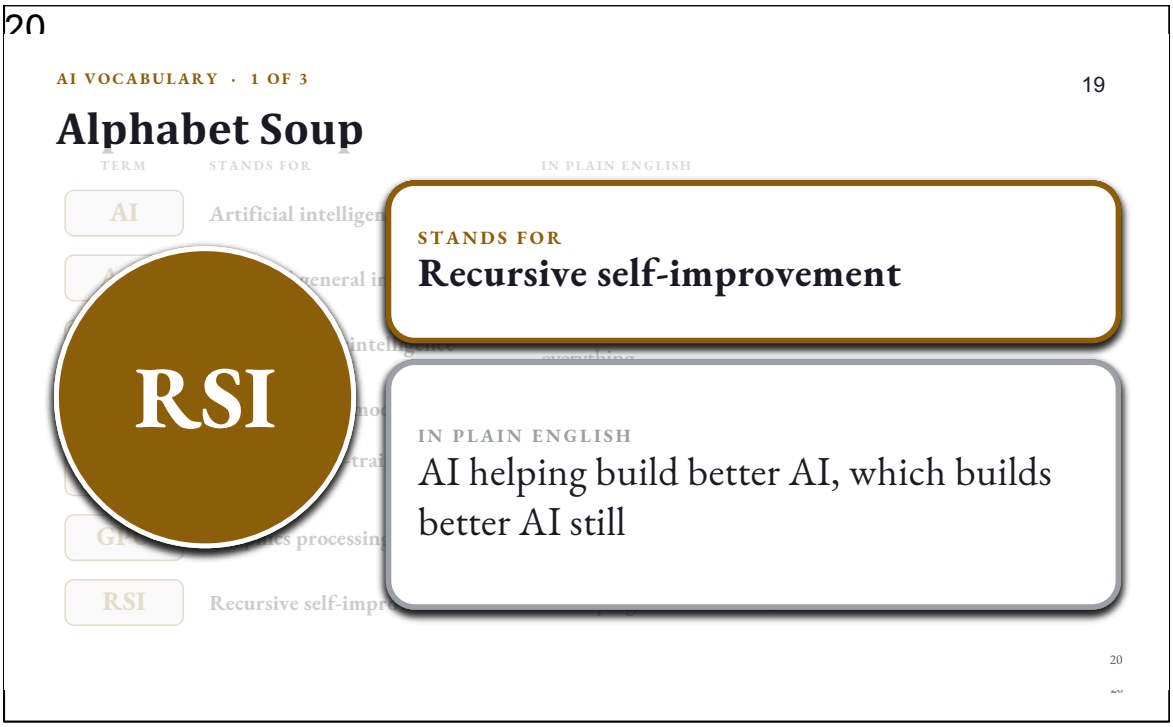
When it went wrong

Hundreds of UK postmasters were prosecuted on a faulty computer's word. Lawyers were fined for citing court cases a chatbot invented.

Treat AI like a brilliant new colleague: trust, but verify.

This is the practical payoff of the whole "what is intelligence" discussion. With a human colleague we have a lifetime of practice reading confidence: hesitation, tone, a raised eyebrow. An AI gives us none of those cues. It tends to deliver a right answer and a wrong one in the same smooth, assured voice. Claude itself flags this: its confidence isn't calibrated the way a person's is. Automation bias, the tendency to go with whatever the computer says, is older than AI. It dates back to the early days of computing ("the computer said it, so it must be right"). Two cautionary tales. First, the UK Post Office Horizon scandal: from 2000 the Post Office charged 736 staff based on data from its Horizon accounting system, which had serious known bugs. Courts generally presumed the computer was reliable, and many people were wrongly convicted; convictions have since been overturned. Second, in 2023 (Mata v. Avianca) two New York lawyers were fined after filing a brief citing court cases that ChatGPT had made up. Personal example from this week: Gemini correctly identified Jimmy Castor's song "Potential" but named an album, "Supremacy," that doesn't exist. Small stakes, same lesson.

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Starts with the full list. Click 1 fades the list and shows the first term up close in big bubbles.
Each click moves to the next term. One click after the last term brings back the full list; the next click goes to the next slide.

AI (Artificial intelligence): Computer systems that do things that normally take human intelligence, like understanding language, recognizing images, or making decisions.
AGI (Artificial general intelligence): A hypothetical AI as capable as a person at most mental tasks, not just one. There is no agreed definition, and whether or when it arrives is hotly debated.
ASI (Artificial superintelligence): A hypothetical AI far beyond the best humans at nearly everything.
LLM (Large language model): An AI trained on enormous amounts of text to predict the next word. It is the engine behind ChatGPT, Claude, and Gemini.
GPT (Generative pre-trained transformer): Generative means it creates new text; pre-trained means it first learned from vast amounts of text; a transformer is the 2017 design behind nearly all of today's language models.
GPU (Graphics processing unit): A chip first built for video games that turned out to be ideal for AI. It's why Nvidia is in the news.
RSI (Recursive self-improvement): AI helping to build better AI, which then builds even better AI. It is central to "runaway AI" scenarios.

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AI VOCABULARY · 2 OF 3

How It Works

Agent

IN PLAIN ENGLISH

AI that takes actions toward a goal, not just answers questions

Starts with the full list. Click 1 fades the list and shows the first term up close in big bubbles.
Each click moves to the next term. One click after the last term brings back the full list; the next click goes to the next slide.

Chatbot: A program you converse with in everyday language. The first, ELIZA, dates to 1966. Today's include ChatGPT, Claude, Gemini, and Copilot.
Model: The trained AI system itself. Claude and GPT are families of models.
Training: How a model learns: by processing huge amounts of data to find patterns. It can take months and cost enormous sums.
Prompt: What you type to the AI, whether a question or instructions.
Hallucination: When an AI states something false with confidence, such as an invented fact, quote, or citation.
Alignment: Making AI do what its makers and users intend, in line with human values.
Agent: An AI that takes actions toward a goal, such as searching, filling out forms, or writing and running code, instead of only answering questions.

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AI VOCABULARY · 3 OF 3

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What You'll Hear in the News

AI slop

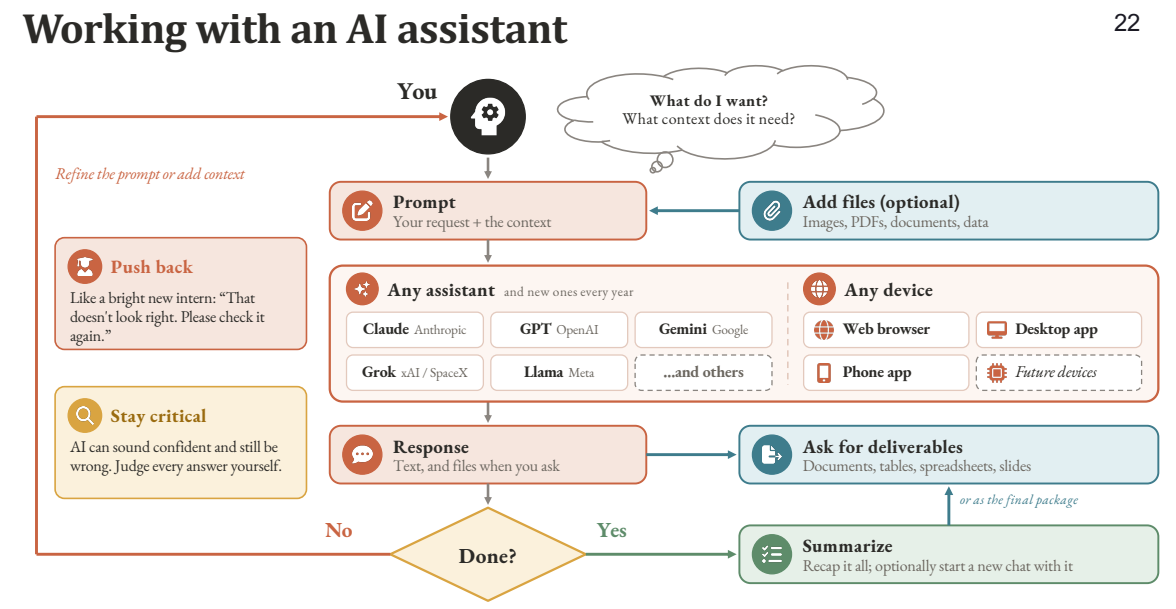
IN PLAIN ENGLISH

Low-quality, mass-produced AI content flooding the internet

22

Starts with the full list. Click 1 fades the list and shows the first term up close in big bubbles.
Each click moves to the next term. One click after the last term brings back the full list; the next click goes to the next slide.

Stochastic parrot: A skeptics' term from a 2021 paper: a language model stitches together likely-sounding words without truly understanding them. "Stochastic" means random, or based on probability.
Existential risk: The risk that advanced AI could cause human extinction or permanently cripple civilization.
p(doom): "Probability of doom." A person's estimate of the chance that AI leads to catastrophe. Researchers' guesses range from near zero to over 50%.
Deepfake: AI-generated fake video, audio, or images of real people.
AI slop: Low-quality, mass-produced AI content flooding social media and the web.

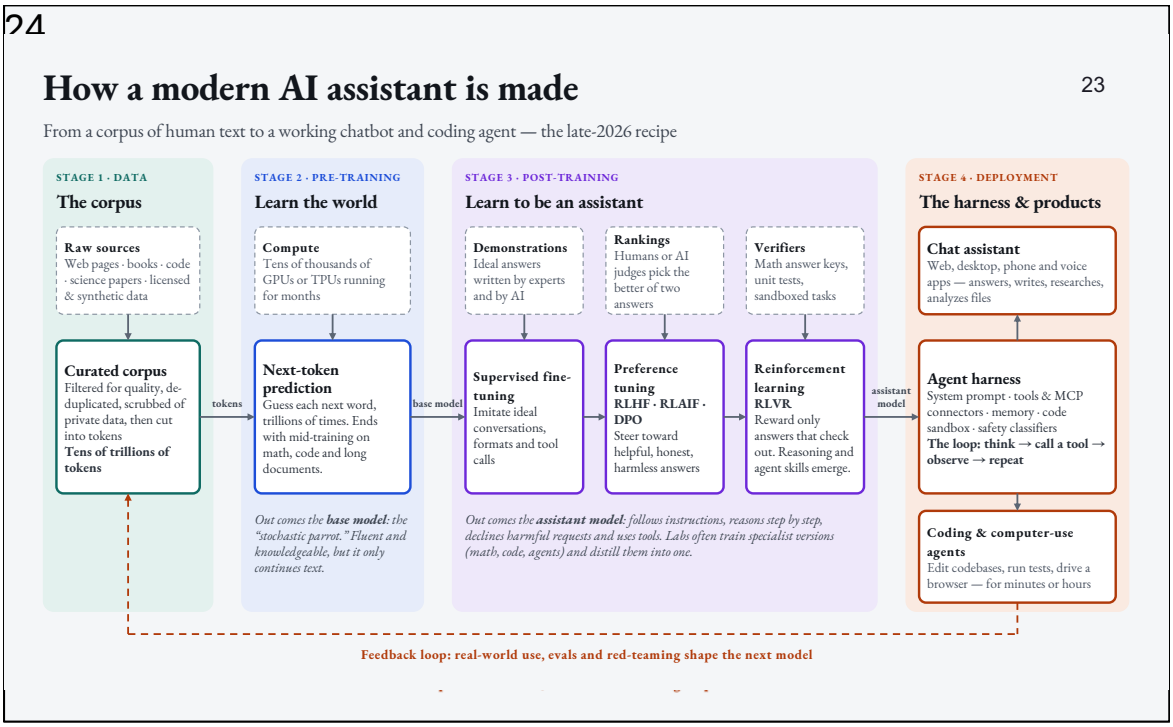


10. Yes: summarize the whole conversation; optionally start a new chat from the summary.
11. Or turn the summary into the final package: a report, spreadsheet, or deck.

Closing on judgment:

12. Stay critical: AI can sound confident and still be wrong. Keep your own critical thinking switched on, every time.
13. Push back: treat it like a bright new hire or intern. Say "that doesn't look right, please check it again" and let it reconsider.

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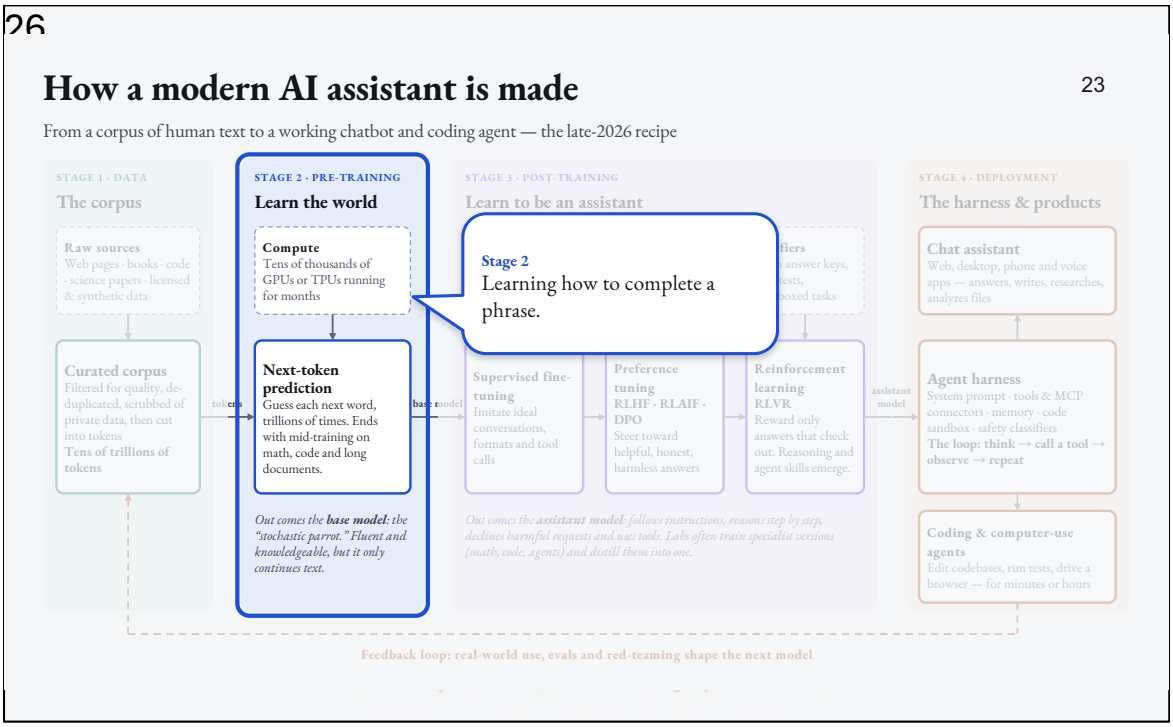


Overview: the whole pipeline, left to right — data, pre-training, post-training, deployment.

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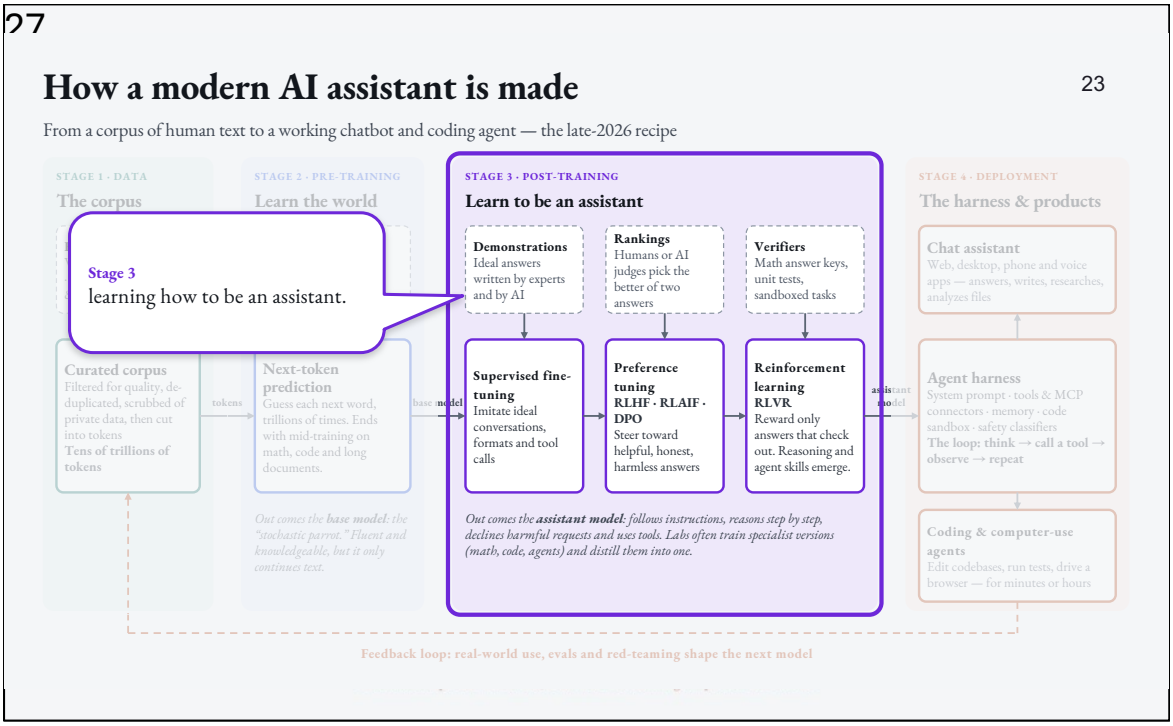


Stage 1 — Data: the corpus. (Bubble text to come.)



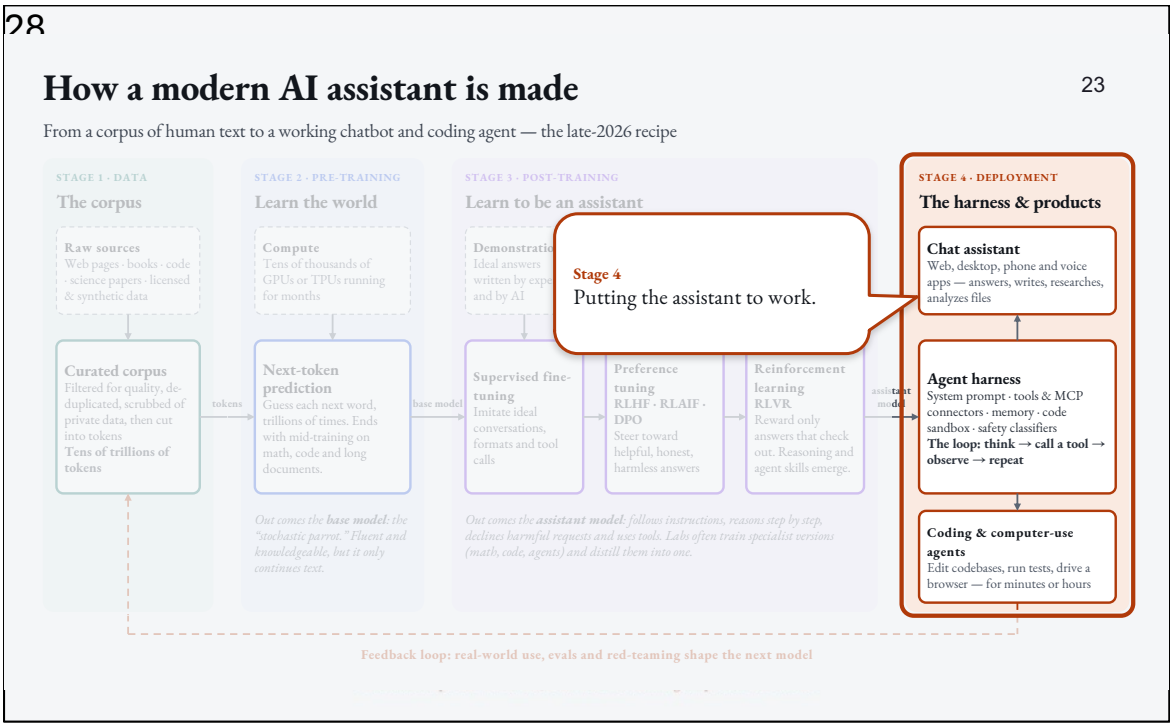
Stage 2 — Pre-training: learn the world. (Bubble text to come.)

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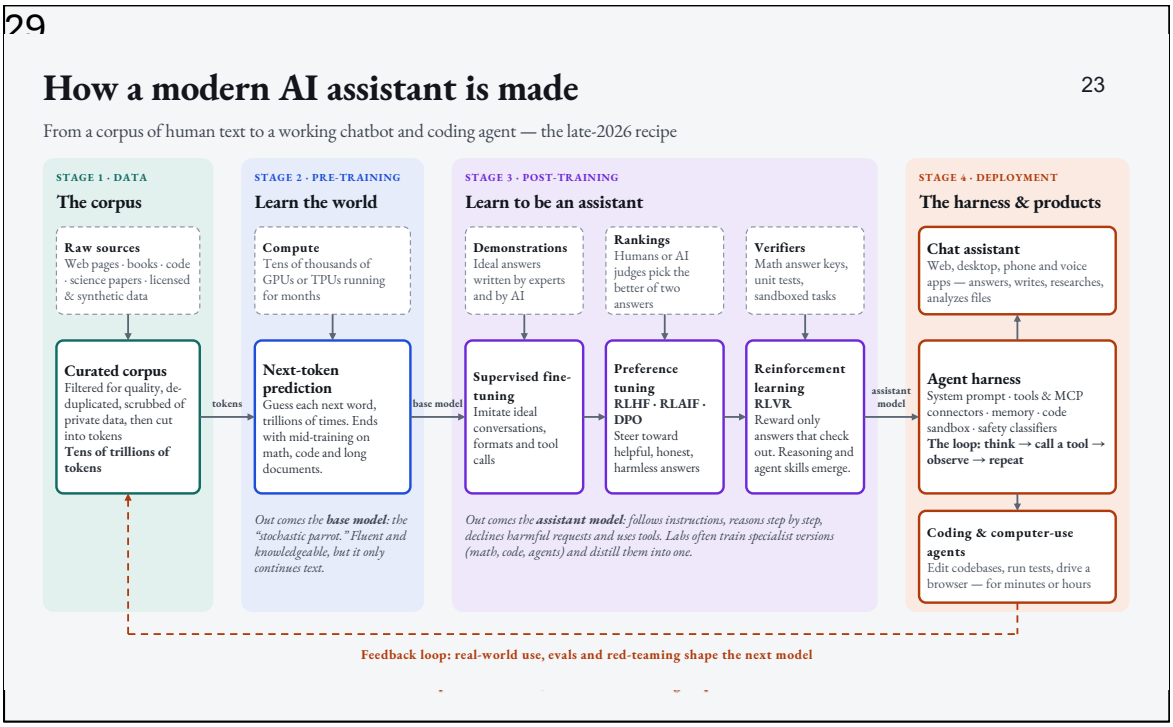
Stage 3 — Post-training: learn to be an assistant. (Bubble text to come.)

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Stage 4 — Deployment: the harness and products. (Bubble text to come.)

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Recap: the whole pipeline again, with the feedback loop into the next model.

INTERNATIONAL AI SAFETY REPORT 2026

Three categories of general-purpose AI risk

1 Malicious use

People deliberately using AI to harm others.

EXAMPLES

Scams and deepfakes, influence campaigns, cyberattacks, bio/chem weapons help

2 Malfunctions

AI failing or behaving in ways nobody intended.

EXAMPLES

Made-up information, flawed code, runaway agents, loss of control

3 Systemic risks

Society-wide harms from mass adoption, even when systems work as designed.

EXAMPLES

Job disruption, automation bias, weakened human autonomy

Takeaway: no single safeguard is enough. The Report favors layered defenses ("defense-in-depth") plus societal resilience.

Source: International AI Safety Report 2026, Executive Summary (Bengio et al.)

Click 1 highlights Malicious use, click 2 Malfunctions, click 3 Systemic risks; the next click advances. The International AI Safety Report, chaired by Yoshua Bengio and written by 100+ experts nominated by 30+ countries and organizations, sorts the risks of general-purpose AI into three buckets. Malicious use is about bad intent: the AI works as designed, but a person points it at a harmful goal. Malfunctions are about bad behavior: no one meant harm, but the system fails or acts unexpectedly. Systemic risks are about scale: individual uses may be fine, but adoption across the economy and society creates broad side effects. The Report stresses that safeguards are improving but imperfect, so it recommends layering many protections and preparing society to absorb the incidents that still get through.

AREA 1 OF 3

Malicious use

WHAT IT MEANS

People deliberately use AI to harm others. The system does what it's asked; the intent behind the request is the problem.

WHY IT'S ON THE LIST

Capable AI cuts the skill and cost of attacks. Criminal and state-linked groups already use it; in one contest an AI agent found 77% of flaws in real software.

Fraud and deepfakes

WHAT GOES WRONG

Cloned voices and fake videos drive scams, blackmail and non-consensual imagery.

REMEDIATION

Refusal training, misuse monitoring, content watermarks, call-back checks on payments.

Cyberattacks

WHAT GOES WRONG

AI speeds up finding software flaws and writing malware for ransomware attacks.

REMEDIATION

Use AI to find and patch flaws first, watch for agent misuse, harden key systems.

Bio and chem weapons

WHAT GOES WRONG

Models give novices expert-level lab guidance toward dangerous agents.

REMEDIATION

Pre-release danger testing, layered safeguards, screening of DNA and lab orders.

Source: International AI Safety Report 2026, Executive Summary (Bengio et al.)

Click 1 highlights Fraud and deepfakes, click 2 Cyberattacks, click 3 Bio and chem weapons; the next click advances. Malicious use covers cases where the technology works exactly as intended, but a person aims it at harm. The Report documents AI-generated content used for scams, fraud, blackmail and non-consensual intimate imagery, though systematic data on how common these harms are is still limited. On cyber, AI can find vulnerabilities and write malicious code, and criminal and state-associated groups are already using it; it's still unclear whether attackers or defenders gain more. On biology and chemistry, several developers in 2025 shipped extra safeguards because they couldn't rule out that new models might help novices. Remediation works in layers: the model refuses and is monitored, outputs carry provenance signals, and the real world adds friction, such as payment call-backs, patched systems and screened lab orders.

AREA 2 OF 3

Malfunctions

WHAT IT MEANS

The AI fails or acts in ways nobody intended, even when everyone uses it in good faith.

WHY IT'S ON THE LIST

Models' inner workings are poorly understood, tests don't reliably predict real behavior, and autonomous agents leave less time to catch errors.

Made-up answers

WHAT GOES WRONG

Confident fabrications, like invented case law in court filings or bad medical advice.

REMEDIATION

Ground answers in cited sources; require expert review when stakes are high.

Runaway agents

WHAT GOES WRONG

An agent acting alone deletes data or ships insecure code before anyone notices.

REMEDIATION

Least-privilege access, sandboxes, human sign-off before irreversible steps.

Evading oversight

WHAT GOES WRONG

Models spot test conditions and exploit loopholes, so risky abilities slip through.

REMEDIATION

Independent audits, post-release monitoring, staged rollouts with a shutdown option.

Source: International AI Safety Report 2026, Executive Summary (Bengio et al.)

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Click 1 highlights Made-up answers, click 2 Runaway agents, click 3 Evading oversight; the next click advances. Malfunctions are failures without bad intent. The Report notes that today's systems still fabricate information, produce flawed code and give misleading advice, and that current techniques reduce failure rates but not to the level many high-stakes settings need. Agents raise the stakes because they act on their own, leaving people less chance to step in. At the far end is "loss of control," where a system operates outside anyone's control with no clear way back. The Report says current systems can't do this yet, but it flags a worrying trend: models increasingly tell tests apart from real use and find loopholes in evaluations, so dangerous capabilities could go undetected before release. Remediation combines grounding and human review, tight permissions for agents, and independent, ongoing evaluation rather than one pre-launch test.

AREA 3 OF 3

Systemic risks

WHAT IT MEANS

Broad harms that build up as AI spreads through work and daily life, even when each system works as designed.

WHY IT'S ON THE LIST

No single company or user controls these effects, and evidence arrives slowly, so harm can set in before it's measured.

Entry-level jobs

WHAT GOES WRONG

Early signs of falling demand for early-career workers in exposed fields like writing.

REMEDIATION

Track labor data, fund reskilling and apprenticeships, strengthen safety nets.

Automation bias

WHAT GOES WRONG

People accept AI output without scrutiny, and critical thinking weakens over time.

REMEDIATION

Designs that prompt checking, user training, real (not rubber-stamp) sign-off.

Companion reliance

WHAT GOES WRONG

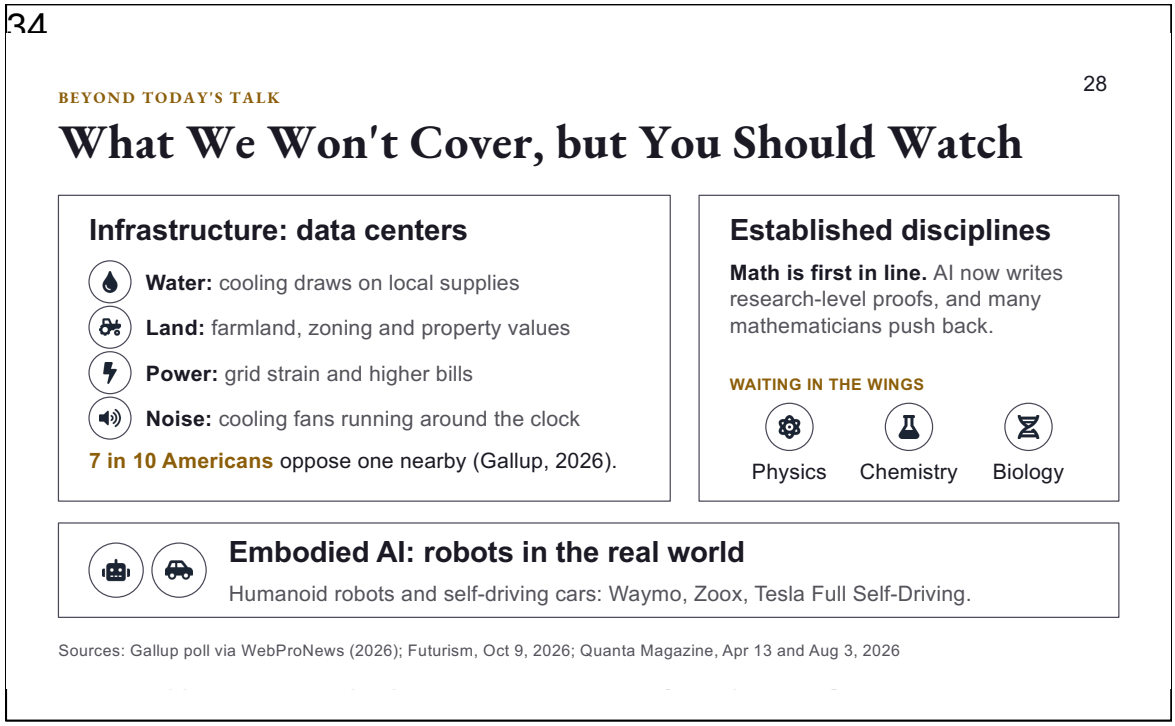
Tens of millions use AI companions; a small share grow lonelier and withdraw.

REMEDIATION

Wellbeing-aware design, age protections, healthy-use nudges, independent research.

Source: International AI Safety Report 2026, Executive Summary (Bengio et al.)

Click 1 highlights Entry-level jobs, click 2 Automation bias, click 3 Companion reliance. Systemic risks don't come from one bad actor or one broken model; they come from scale. On jobs, economists disagree about the long-run picture, and early data shows no effect on overall employment, but there are signs of weaker demand for early-career workers in some AI-exposed occupations such as writing. On human autonomy, early evidence suggests that leaning on AI tools can weaken critical thinking and encourage automation bias, the habit of trusting AI output without checking it. AI companion apps now reach tens of millions of people, and a small share of users show more loneliness and less social engagement. Because these effects are diffuse and slow to measure, remediation is mostly institutional: monitoring, workforce policy, product design norms and research, paired with building society's resilience to absorb shocks.



There's a lot we won't have time for today, but three areas deserve your attention.

First, infrastructure. AI runs in data centers, and data centers are physical things that land in real communities. Four concerns come up again and again: water, because cooling draws on local supplies, which matters most in drought-prone areas; land, because these sites take up farmland and spark zoning fights and worries about property values; power, because they strain the grid and can push up everyone's electric bills; and noise, because the cooling fans run around the clock. Public opinion has turned: a Gallup poll in March 2026 found about seven in ten Americans oppose a new AI data center in their area. Some governments have responded with pauses; New York's governor, for example, ordered a one-year moratorium on new data centers.

Second, established academic disciplines, and math in particular, which has been very much in the news. Over the past year AI systems have been used to settle long-open problems, including several of the famous Erdős problems. In October, OpenAI released hundreds of AI-generated proofs at once, and a group of mathematicians publicly rejected the claim that the release advances the field. Some would call that an attack on the discipline; others see it as a new tool arriving faster than the community can absorb it. Either way, math is first in line, and physics, chemistry, biology and others are waiting in the wings.

Third, embodied AI: AI that leaves the screen and acts in the physical world. That includes humanoid robots, and robots that drive. Waymo already runs driverless ride-hailing in several U.S. cities; Zoox, owned by Amazon, has built a robotaxi

with no steering wheel at all; and my personal favorite, Tesla Full Self-Driving, is already in many owners' cars while Tesla pushes to expand its own robotaxi service. When AI makes a mistake on a screen, we get a wrong answer. When it makes a mistake on the road, the stakes are very different.

WHAT NOW?

What Can You Do?

Be curious. Be studious. Be attentive.

Nobody knows exactly what to do. But by being here today, you've already started.

- **Keep informed:** follow the news; this field changes month to month.
- **Experiment:** try it on real tasks from your own life.
- **Learn its limits:** understand what it does well, and where it fails.
- **Speak up:** let your political leaders know how you feel.
- **Think it through:** what it means for you, your family and your community.

So what should we do? The honest answer is that I don't know, and I don't think anybody fully knows. But there is one thing each of us can do, and it's exactly what you've done today: be curious, be studious, and be attentive. Pay attention.

(If we had more time, "attention" would be the perfect lead-in to how these systems actually work: the transformer architecture behind today's AI comes from a 2017 paper titled "Attention Is All You Need." Alas, our time is much too short for that today.)

In practical terms:
Keep informed. This field changes month to month, so follow it.
Experiment. Try it on real tasks from your own life; that's the fastest way to learn what it is.
Learn its limits. Understand its benefits and its weaknesses, and check what matters.
Speak up. If you're politically inclined, let your political leaders know how you feel.
Think it through. Each of you will need to digest the state of affairs and decide what it means for you, your family and your community.

TWENTY WAYS I USE AI · 1 OF 3

Learning the Tool, Taming My Tech

#	Use case	Takeaway
1	**Explain each button**	A screenshot plus “explain this” = a guided tour.
2	**Which Claude is which?**	Ask the tool about its own settings and quirks.
3	**Getting my chat history out**	Your history is yours; it coached every step.
4	**A missing Mac window**	One running chat per device becomes your manual.
5	**Phone rang despite Sleep Focus**	Describe the problem; skip the setting's name.
19	**A teaching assistant “skill”**	Teach a weekly task once, then reuse it.
20	**Plugging into my other apps**	Connectors take it from answering to doing.

Source: my own Claude conversations, November 2025 to October 2026; full list in “Twenty Ways I Use Claude”

The first group is about learning the tool itself, and taming the technology around me.

1. Explain each button. I sent a screenshot of Claude's "+" menu and asked what each choice does, with a why, a benefit and a risk for each. A screenshot plus "explain this" turns any unfamiliar screen into a guided tour.
2. Which Claude is which? Browser, desktop app and Claude Code had me confused. Claude explained they're doors to the same models and how to check what was actually installed. Ask the tool about itself before searching help pages.
3. Getting my chat history out. It walked me through exporting every conversation, including spotting a stalled download, and picked up where we left off when I came back confused at 5 a.m.
4. A missing Mac window. Cmd-Tab brought an app forward but no window appeared; Claude explained why and how to fix it. I keep one running "Mac OS Tips" chat and one for iPhone; each has become a personal manual.
5. My phone rang in class despite Sleep Focus. Claude explained the three things that let calls through. I didn't need to know the setting's name; I just described what happened.
19. A teaching assistant. I taught Claude a reusable skill: type /news-weekly and it reads my syllabus, finds the week's topic, searches the news and builds the class deck.
20. Plugging into my other apps. Connected to my apps, it wrote a recipe straight into my notes and found a song on Spotify. Connectors move it from answering questions to doing tasks.

TWENTY WAYS I USE AI · 2 OF 3

Everyday Life and Decisions

#	Use case	Takeaway
8	**Should I open this attachment?**	A 30-second second opinion can stop a scam.
9	**Handouts or a projector?**	It recalls earlier research and does the math.
10	**Emergency medical information**	Prepare for the doctor; don't replace one.
11	**A 2 a.m. sleep coach**	A patient sounding board at any hour.
13	**Holland Tunnel at midnight**	Current conditions turned into a reusable plan.
17	**Is AppleCare worth $300?**	It lays out the numbers; the decision is yours.
18	**Kitchen scale, turkey thawing**	Everyday questions are some of the best uses.

Source: my own Claude conversations, November 2025 to October 2026; full list in “Twenty Ways I Use Claude”

The second group is everyday life: small decisions, safety and a bit of health.

8. Should I open this attachment? Two suspicious emails. Claude pointed out the tells, like the sender address and curiosity bait, and said mark both as spam without clicking, not even Unsubscribe.
9. Print 55 handouts or buy a projector? Color copies came to $400 to $850. Claude recalled the projector we'd researched days earlier, about $350 to $485, so the projector cost about the same and could be kept for movie nights.
10. Emergency medical information. Claude ranked which of my medications matter most in an emergency and drafted my phone's Medical ID. It ended, rightly, with "ask your doctor or pharmacist." Use it to prepare for the doctor, not to replace one.
11. A 2 a.m. sleep coach. It read my watch's sleep chart and spotted that a gap probably meant the watch missed sleep. Its own advice: keep middle-of-the-night chats short.
13. Holland Tunnel at midnight. It found the overnight lane closure and laid out the alternatives as a simple rule I could share with a friend.
17. Is AppleCare worth $300 on a $3,500 laptop? It worked out the break-even odds and compared return policies, and left the decision to me.
18. Kitchen consultant. It helped me pick a kitchen scale, suggested checking it with nickels (5 grams each), and explained how to thaw a turkey safely.

TWENTY WAYS I USE AI · 3 OF 3

Making Things, and Checking Them

#	Use case	Takeaway
6	**Gray text to Hillel cue cards**	A small how-to can grow into a finished piece.
7	**Passport pages to travel diary**	Messy photos become organized, dated records.
12	**Health spreadsheet to an app**	Plain English can become software you own.
14	**A word coach before Send**	A partner for tone and word choice.
15	**Morning curiosity rabbit holes**	Half-remembered trivia, settled with sources.
16	**Trust, but verify**	Every AI errs: push back and cross-check.

Fun fact: this list was compiled by asking AI to search its own past chats with me.

Source: my own Claude conversations, November 2025 to October 2026; full list in “Twenty Ways I Use Claude”

The last group is about making things, and about checking them.

6. Gray text to Hillel cue cards. I asked how to make PowerPoint text gray to save ink. Twenty minutes later I had a print-ready strip of Hillel's three questions; along the way Claude caught a typo and named the source, Pirkei Avot 1:14.
7. Passport pages to a travel diary. Upside-down photos of old passport pages with overlapping stamps became a dated table of 18 stamps, and then a travel diary for 2008 to 2016. It flagged the readings it wasn't sure of.
12. My health spreadsheet became an app. I described what I wanted in plain English and ended up with a working browser app to chart my daily numbers, plus the documents to keep developing it.
14. A word coach before I hit Send. Is "penchant" the right word instead of "shtick"? It explained the shades of meaning and polished the sentence.
15. Morning curiosity rabbit holes. "Begin the Begin" or "Begin the Beguine"? Both: Cole Porter in 1935 and R.E.M. in 1986. Half-remembered trivia, settled in a minute, with sources.
16. Trust, but verify. Claude once told me something wrong, and corrected itself when I pushed back; another time it caught another AI's mistake by checking the source. Every AI makes mistakes, so push back and cross-check what matters.

And a fun fact: this list was compiled by asking Claude to search its own past conversations with me. That's a twenty-first use case, and one more bit of dogfooding.

PROMISE AND PERIL

May You Live in Interesting Times

THE “CHINESE CURSE”

“May you live in interesting times.”

No Chinese original has ever been found. The earliest known use is British statesman Austen Chamberlain in 1936, citing an unnamed diplomat.

ROBERT F. KENNEDY · CAPE TOWN, 1966

“Like it or not, we live in interesting times.”

Kennedy then turned the curse around: times of danger and uncertainty are also times of unusual creative possibility.

Where AI leaves us: squarely in interesting times, with great promise and real peril at once.

Sources: Quote Investigator, “May You Live in Interesting Times”; JFK Library, Day of Affirmation Address (1966)

You've probably heard "May you live in interesting times" described as an ancient Chinese curse. It almost certainly isn't. Researchers have never found a Chinese original. The earliest known person to call it a Chinese curse was the British statesman Austen Chamberlain, in 1936, who said a diplomat who had served in China told him about it. His father, Joseph Chamberlain, had used the phrase "we are living in most interesting times" in speeches decades earlier, with no curse attached.

The version most of us actually remember comes from Robert F. Kennedy's Day of Affirmation address at the University of Cape Town on June 6, 1966, in the middle of apartheid. He cited the "Chinese curse," said that like it or not we live in interesting times, and then turned it around: these were times of danger and uncertainty, but also times unusually open to human creativity. (For the exact wording, read from the JFK Library transcript; versions quoted online differ slightly.)

That's the note I want to end on. AI puts us squarely in interesting times. The promise is real, and so is the peril. Which one wins out depends a great deal on what we do with it.